



Department of Computer Science

Assignment 2

Course Code: CS-201

Course Title: Data Structures

Total Marks: 15

Date: 21-11-2025

Name: _____

Registration No. _____

Discipline/Semester: BSCS- 3rd A

Submission deadline: 26-11-2025

Instructions:

- Late submissions will not be entertained.
- Copied assignments will be marked ZERO.
- Use of AI tool is strictly not allowed.
- Submit the assignment in hard form on A4 pages.

CLO2, GA4, C3

QUESTION

As a software developer you are required to develop a Library Management System that manages the library's book catalog, where each book is represented as a node in a doubly linked list. Each node contains key information i.e. book ISBN. This structure allows the library to maintain the order in which books were added while providing the flexibility to easily remove books when they are no longer part of the collection. Your task is to **construct** an algorithm to add a feature to delete books from this catalog, covering three specific cases by ensuring that links between nodes are appropriately maintained and that each deletion scenario is handled effectively: deleting the first book in the catalog, deleting the last book, and deleting a specific book by its ISBN, which could be located anywhere within the catalog.

Note: Also show the pictorial representation of each deletion case.